Node Swap

AlgoExpert · Very Hard · Linked Lists

The problem

- Swap every pair of adjacent nodes (not values); return the new head.

Intuition

- Swap pairs by rewiring three links at a time, using a dummy head.

Brute force vs optimal

- Swap values → not allowed
- Recursion → O(n) stack
- Iterative with dummy → O(1)

Approach

- 1. Dummy before head
- 2. For each pair, rewire previous, first and second
- 3. Advance by two

Worked example



After 0 swaps

Worked example



After 1 swaps

Worked example



After 2 swaps

Worked example



After 3 swaps

Java solution

```java
public class NodeSwap {
    public static LinkedListNode nodeSwap(LinkedListNode head) {
        LinkedListNode dummy = new LinkedListNode(0);
        dummy.next = head;
        LinkedListNode p = dummy;
        while (p.next != null && p.next.next != null) {
            LinkedListNode a = p.next, b = a.next;
            a.next = b.next; b.next = a; p.next = b;
            p = a;
        }
        return dummy.next;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Odd length leaves last node
- Return dummy.next

Interview tips & follow-ups

- Follow-up: reverse in k-groups

Recap

- Swap pairs by rewiring three links at a time, using a dummy head.
- Complexity: O(n)